COBBLE AWS PLATFORM

Secure foundation. Rebuilt from code.

A multi-account AWS landing zone and proof of concept for the Cyber Security Centre



Team T235

Industry partner: Cyber Security Centre

JH **Jacob Hardcastle**
Definition

TH **Tim Hartmann**
Design

MG **Max Gosling**
Demo and plans

KD **Khai Do**
Operations

DEFINITION

What is Cobble?

A proof-of-concept Software-as-a-Service platform from the Cyber Security Centre: many customer organisations (tenants) share one platform, built from modules that each deploy on their own.

/apl

Application Plane

What users sign in to

- React application shell
- Login and identity context
- Loads each module's screen

/adm

Administrative Plane

Runs the platform itself

- Tenancies
- Module management: install, version, disable
- Logging and audit (LOG)

/aum

Application User Modules

What tenants actually use

- Member management (MMS) first
- More modules over time
- e.g. 20 modules = 20 services

Every module owns
its screen, its API, its data and its release

Tenants kept apart by
row-level security in SQL Server

One faulty module
can be disabled or rolled back alone

We don't build the modules: we build the secure AWS foundation they run on

DEFINITION

Two solutions. One set of building blocks.

B Minimum viable product

Working today

≈ A$0

- One account, 48 resources
- Encryption, audit trail, threat detection, budget alert
- Torn down between demos

A Full landing zone

What the Cyber Security Centre would run

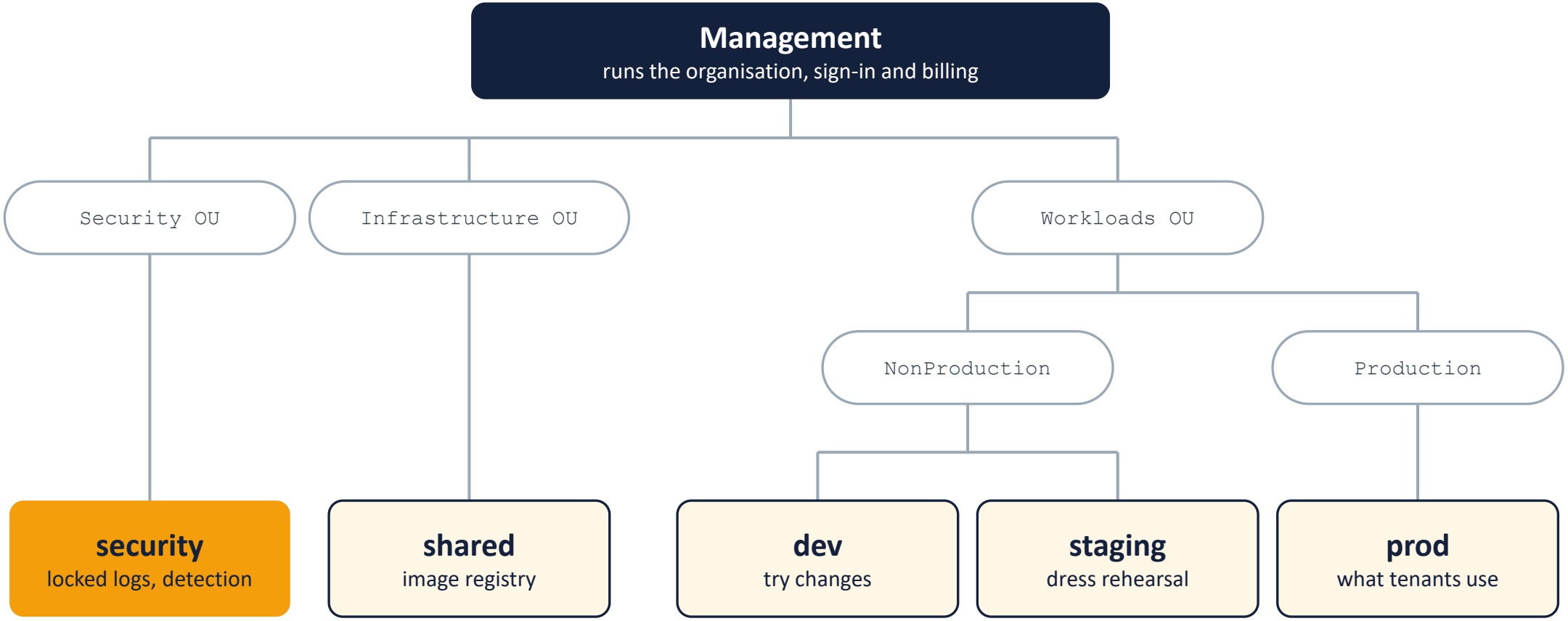
≈ A$790 / month

- Six accounts with guardrails and single sign-on
- Locked central logs and security monitoring
- dev, staging and prod from one workload folder

Same modules: B is a working slice of A, not a throwaway prototype

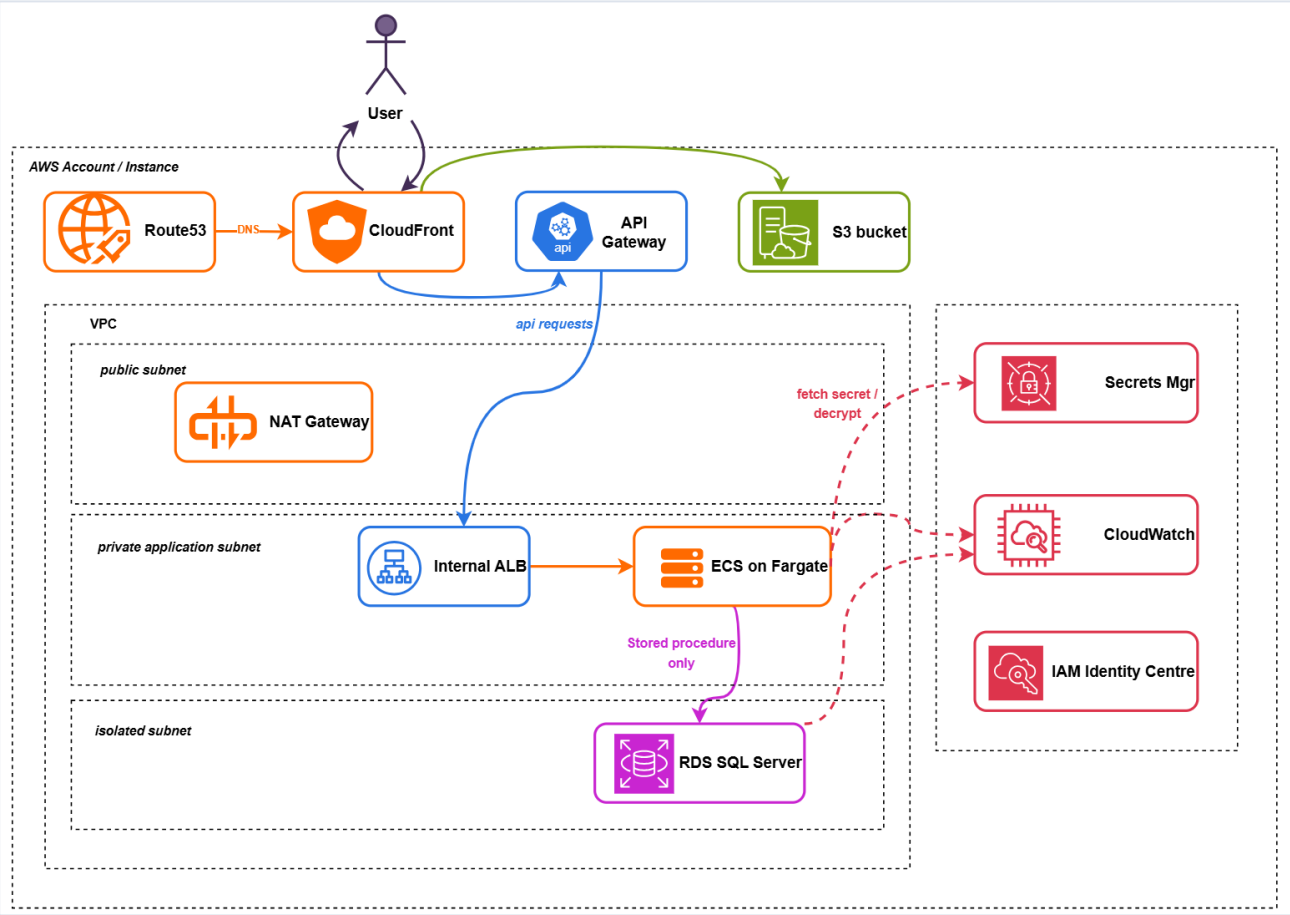
OUTPUTS

Six accounts, each with one job



OUTPUTS

One request, from the internet to the database



1 Edge

Route 53 and CloudFront with a web firewall. Static pages from a private bucket

2 API

API Gateway to an internal load balancer, never public

3 Containers

ECS on Fargate in private subnets; secrets fetched at start-up

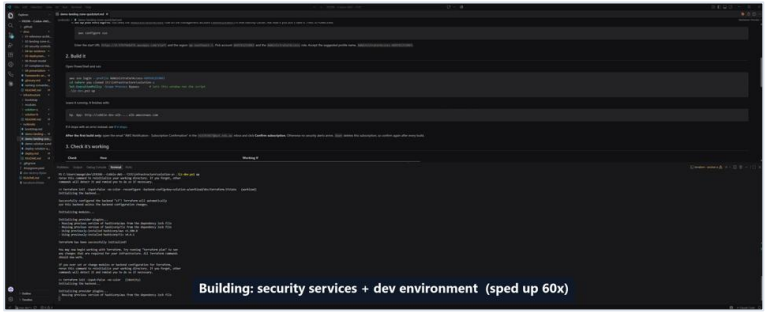
4 Database

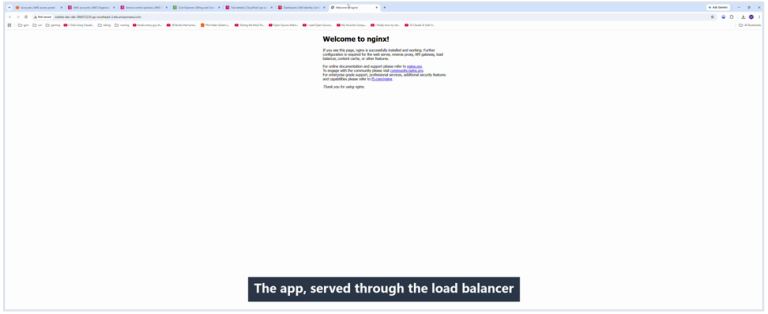
SQL Server in an isolated subnet, stored procedures only, no route out

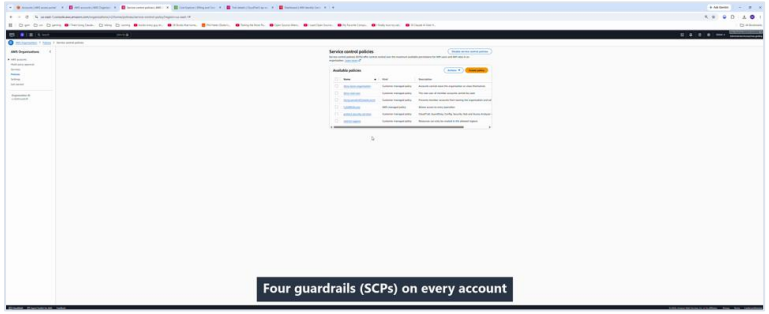
Target from the Implementation Guide §2. Built today: load balancer, private containers, private encrypted database. Edge and API Gateway follow once there is a domain (Phase 2).

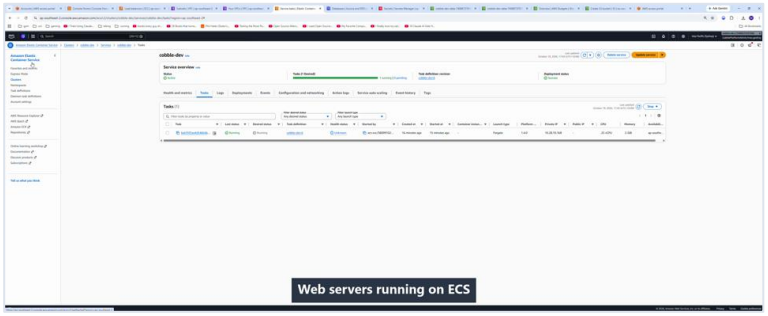
OUTPUTS AND PLANS

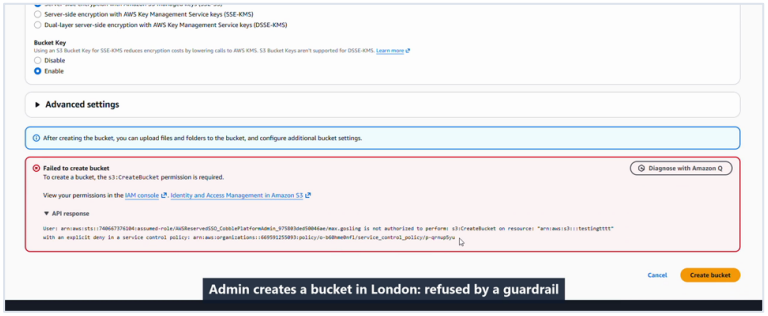
Built, checked and torn down from code: key frames

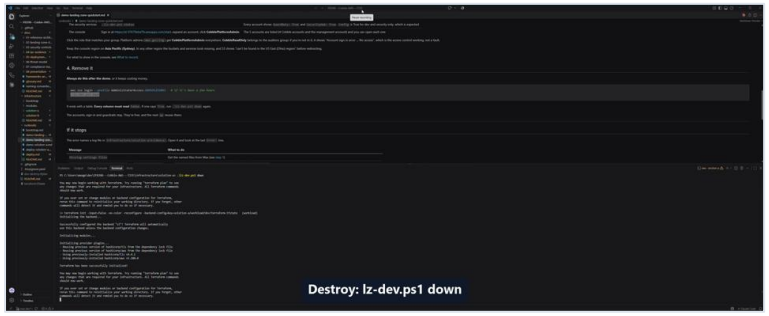










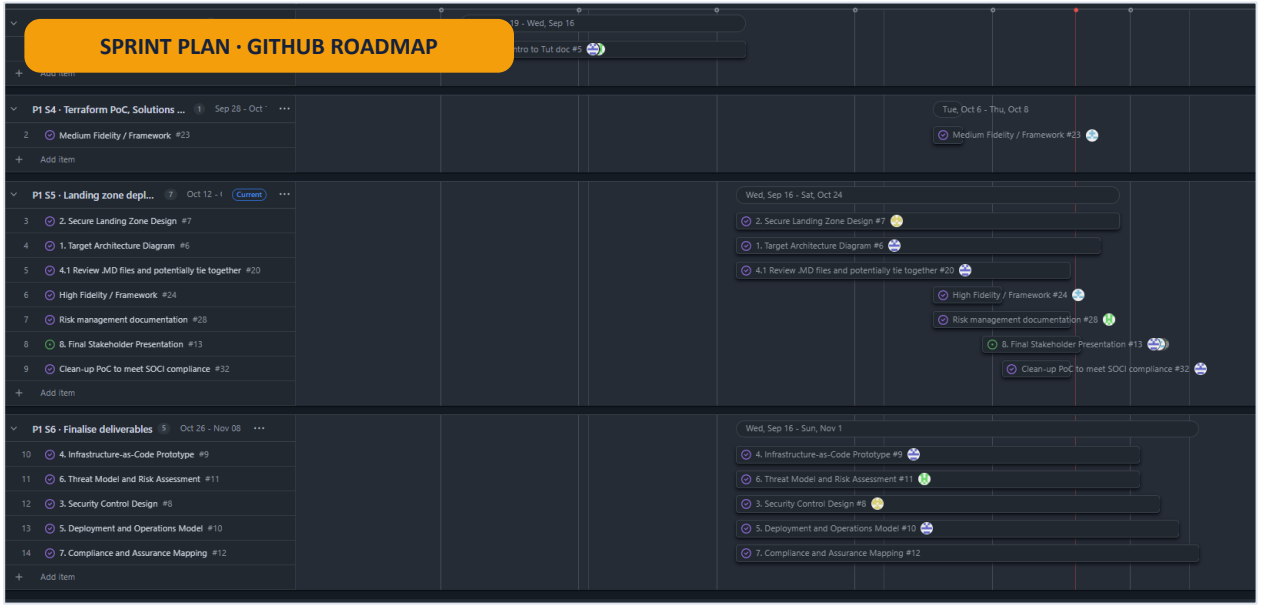


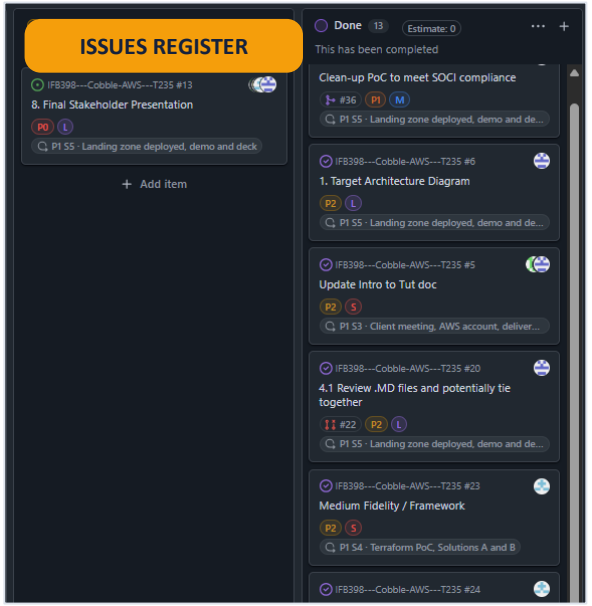
Recorded run, 18 October 2026, cut to two minutes and narrated live: build, the app behind the load balancer, guardrails, private web servers, an admin refused outside Australia, and one-command teardown.

OUTPUTS AND PLANS

How we planned Phase 1

S1 · 17 Aug	S2 · 31 Aug	S3 · 14 Sep	S4 · 28 Sep	S5 · 12 Oct	S6 · 26 Oct
Repo and workflow	Research, pricing	Client meeting, scope	Terraform PoC, A and B	Landing zone, demo, deck	Finalise deliverables





6
two-week sprints

13 / 14
deliverable issues done

19
pull requests merged

Sprint plan: each deliverable an issue with owner, priority, size and dates, in its sprint. **Issues register:** 13 done, the final presentation in progress.

Sprint plan and issues register in GitHub · every change a pull request, checked and reviewed

OUTPUTS AND PLANS

Phase 2 is the Centre’s decision to make

Phase 1 gives them three ways to configure Cobble’s platform. They choose and sign off; then we build.

OPTION 1

Full landing zone

≈ A$790 / month

- Solution A in the Centre’s own AWS organisation
- Six accounts, guardrails, single sign-on
- A$475 with dev and staging off when idle

Needs: their AWS organisation, budget and identity provider

OPTION 2

Start small, grow

≈ A$0 between demos

- Solution B in one Centre account
- Grows into Solution A when tenants arrive
- Same modules, so nothing is thrown away

Needs: one AWS account

OPTION 3 NOT YET APPROVED

Review DigitalOcean first

Evidence before building

- Map today’s platform against the ten control areas
- Then choose Option 1 or 2 with the evidence
- Migration or hardening plan

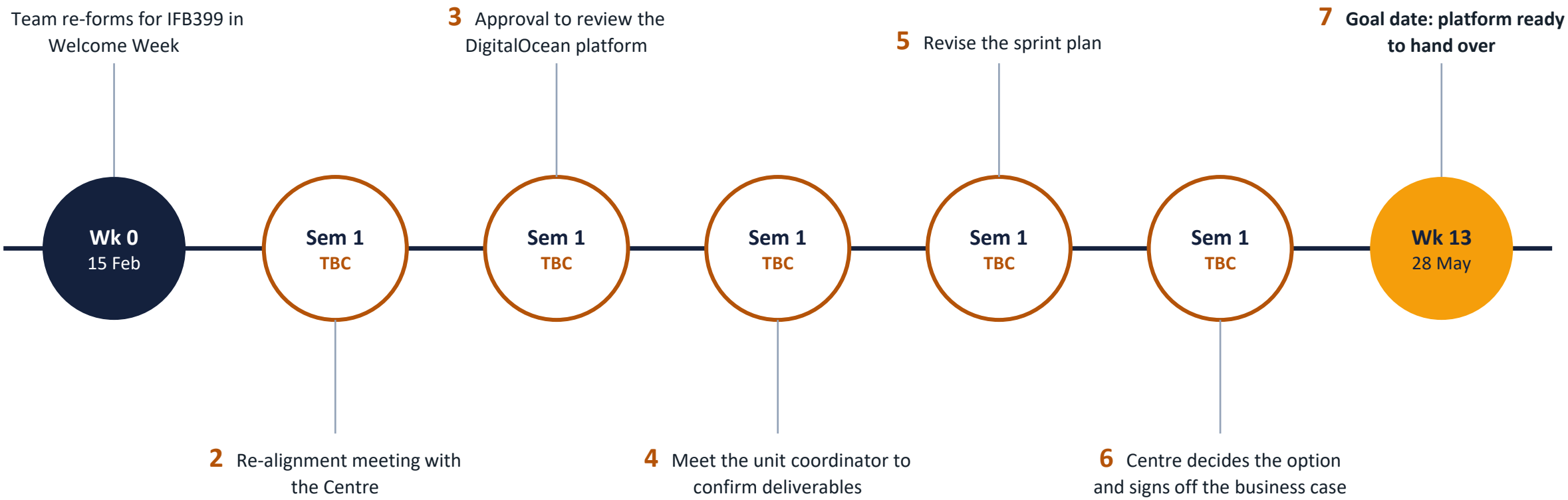
Needs: development approval to review DigitalOcean

First: re-confirm scope with the Centre · Then: they choose an option and sign off the plan and business case

OUTPUTS AND PLANS

Going forward in IFB399

TBC: depends on the Centre and the unit



QUT Semester 1 2027: Welcome Week 15 Feb · teaching ends 28 May · exams 5–19 June

OUTPUTS AND PLANS

Why this plan is feasible

01

It’s already running

Both solutions deploy and tear down from code. Phase 2 extends a working platform.

02

The work is named and ordered

13 threats rated; each of the 3 High ones has a named fix.

03

The decision can’t stall the build

The same modules serve whichever option the Centre picks; only the target accounts change.

RISKS TO THE PLAN, RATED BY IMPORTANCE

DigitalOcean review access not approved — VERY IMPORTANT

Ask in writing at re-alignment; if no answer, keep building

The Centre’s decision takes time — VERY IMPORTANT

The same modules serve every option

No Centre accounts in time — IMPORTANT

Handover includes a step-by-step deploy guide

Account limits and cost — IMPORTANT

Keep accounts; tear down only what costs money

Exams squeeze the end — IMPORTANT

Goal date is the end of teaching, 28 May

Phase 2 isn’t a restart: it builds on the platform you’ve just watched deploy

OPERATION

Three risks to fix before real tenant data arrives

HIGH · T1

Tenant isolation

Tenants are kept apart only by the application

Fix: a database user and role per module

HIGH · T2

Admin access

One stolen admin session can change anything

Fix: narrower roles and a break-glass role

HIGH · T9

Backups

No copy outside the account they protect

Fix: AWS Backup to another account and region

13 threats rated with STRIDE: the other ten are Medium or Low, acceptable for a proof of concept

OPERATION

Team members

JH

Jacob Hardcastle

IFB398

Definition lead

Brief, requirements, scope, client meetings

IFB399 · PROPOSED

Client liaison

Re-alignment, option and business case sign-off

TH

Tim Hartmann

IFB398

Design lead

Landing zone, security controls, risk register

IFB399 · PROPOSED

Security lead

Admin roles, break-glass, guardrails, threat re-rating

MG

Max Gosling

IFB398

Project manager, build lead

Terraform, demo, sprint plan and issues register

IFB399 · PROPOSED

Project manager, platform build

Sprint plan, deploy pipeline, DigitalOcean review

KD

Khai Do

IFB398

Operations lead

Deployment, monitoring, backup, support model

IFB399 · PROPOSED

Operations lead

Monitoring, backups and restore test, handover

Each of us leads one part now, and owns one part of Phase 2